

MASS-APP-014-V15 — Cross-Application Context Assembly & Enterprise Synthesis

Document Control

Field	Value
Document ID	MASS-APP-014-V15
Version	1.0
Status	Production Baseline v1.0
Authority	EWO-MASS-APP-014-V15
Date	2026-08-03

1. Purpose

V15 assembles authorized references from multiple MASS applications, engines, and governed integrations into a traceable working picture for human decisions. It reduces fragmented understanding without becoming a shadow system of record, bypassing source permissions, or converting uncertainty into truth.

2. Permanent Architecture

V15 owns context requests, assembly plans, eligible-source decisions, authorization evidence, reference resolution, context snapshots, contradiction reports, completeness and confidence indicators, redaction records, synthesis packets, and retirement. Source applications own their records; ENG-009 owns AI orchestration; ENG-024 owns analytics; ENG-027 owns lineage.

Every assembly is point-in-time, cited, tenant-scoped, purpose-bound, and immutable after issuance. Refresh creates a new snapshot. Missing, stale, forbidden, and contradictory sources remain visible as metadata rather than silently omitted.

3. Role Mapping

Role	Baseline Mapping	Authority
Context Viewer	Viewer specialization	Read authorized assemblies
Context Requester	Contributor specialization	Request purpose-bound context
Context Steward	Steward specialization	Govern plans, quality, redaction, and issuance
Source Liaison	Contributor extension	Resolve source metadata without granting access
Synthesis Approver	Administrator specialization	Issue executive synthesis packets
Context Administrator	Administrator specialization	Configure gateways and tenant policy

These roles specialize APP-014 roles. Requesters cannot approve their own executive synthesis.

4. Platform Consumption Map

Source	Consumption	Ownership Preserved
--------	-------------	---------------------

APP-001–APP-014 | Authorized record and capability references | Each application remains system of record

ENG-003/004/005 | Identity, authorization, events | Platform-owned

ENG-007/008/009 | Knowledge, documents, synthesis | Platform-owned

ENG-011/015 | Audit and APIs | Platform-owned

ENG-024/027 | Analytics, lineage, quality | Capability-owned

Approved integrations | Governed external references | External source remains authoritative

Publishes context.requested, context.partial, context.contradiction.detected, context.snapshot.issued, context.retired.

5. Gateway Inventory

Gateway | Contract | Failure Behavior

IdentityGateway / AuthorizationGateway | ENG-003/004 principal and decision | Deny source access; preserve denial metadata

ApplicationContextGateway | APP-001–014 read contracts | Mark unavailable or stale source

KnowledgeGateway / DocumentGateway | ENG-007/008 references | Partial assembly with cited gap

AIOrchestrationGateway | ENG-009 synthesis | Return unsynthesized evidence packet

AnalyticsGateway | ENG-024 indicators | Exclude metric and reduce completeness

LineageGateway | ENG-027 lineage validation | Block issuance when mandatory lineage fails

IntegrationGateway | Approved external adapters | No direct connector ownership in V15

EventGateway / AuditGateway | ENG-005/011 | Outbox and retry-safe audit delivery

6. Context Request Lifecycle

Requested → Purpose Authorized → Sources Discovered → Eligibility Evaluated → Assembly Planned → References Resolved → Contradictions Analyzed → Redacted → Synthesized → Human Reviewed → Issued → Retired

The request records purpose, audience, subject, time horizon, required source classes, requested freshness, and permitted sensitivity. Purpose limitation is reevaluated on refresh.

7. Assembly, Freshness, and Contradiction

Source eligibility requires tenant match, principal authorization, valid purpose, active gateway, and acceptable source lifecycle. Each included reference records source type, identifier, version, observed time, freshness policy, permission decision, lineage, and citation locator.

Completeness reports eligible, included, missing, forbidden, stale, failed, and contradictory source counts. Confidence is an explainable synthesis indicator based on source quality, freshness, coverage, and contradiction. It is not factual certification.

Contradictions preserve both claims, sources, timing, and scope. V15 may explain differences but only an authorized human or owning source can resolve them.

8. Snapshots and Synthesis Views

Snapshots are immutable point-in-time manifests of references and indicators. Executive views emphasize enterprise state, decisions required, risks, opportunities, and missing evidence. Operational views emphasize current context, dependencies, constraints, and source freshness. Both remain advisory.

NOVA synthesis and POPS historical continuity are advisory request types executed through ENG-009 and must cite the snapshot evidence.

9. Data Model

Entity | Purpose | Immutable Condition

ContextRequest | Purpose-bound request | After authorization

ContextSourceProfile | Governed source contract | Versioned after activation

ContextAssemblyPlan | Required and optional source plan | After execution begins

SourceEligibilityDecision | Authorization and purpose result | Always

ContextReference | Resolved source reference | After snapshot issuance

ContextSnapshot | Point-in-time assembly | After issuance

ContextCitation | Evidence locator and lineage | After issuance

ContextContradiction | Conflicting claims | After disposition

ContextRedaction | Restricted-field action | Always

ContextIndicator | Completeness/confidence/freshness | After issuance

SynthesisPacket | Executive or operational view | After issuance

ContextAdvisoryRequest | NOVA/POPS request | After completion

ContextOutbox | Reliable events | After publication

10. API Contracts

Method | Path | Purpose | Role

POST/GET | /context-requests | Request or browse context | Requester/Viewer

GET | /context-requests/{id} | Read request status | Context Viewer

POST | /context-requests/{id}/authorize | Authorize purpose | Context Steward

POST | /context-requests/{id}/assemble | Execute assembly plan | Context Steward

GET | /context-snapshots/{id} | Read snapshot | Context Viewer

POST | /context-snapshots/{id}/refresh | Create refreshed snapshot | Context Requester

GET | /context-snapshots/{id}/citations | Read citations | Context Viewer

GET | /context-snapshots/{id}/contradictions | Read contradictions | Context Viewer

POST | /context-contradictions/{id}/disposition | Record human disposition | Synthesis Approver

POST | /context-snapshots/{id}/issue | Issue immutable snapshot | Context Steward

POST | /synthesis-packets/{id}/issue | Issue synthesis packet | Synthesis Approver

POST | /context-advisory-requests | Request NOVA/POPS synthesis | Context Steward

11. Security and Failure Behavior

RLS uses tenant identity from auth.jwt(). Source authorization is checked during discovery, resolution, snapshot read, and citation expansion. Restricted fields are redacted before persistence in a synthesis packet. Immutable snapshots, citations, eligibility decisions, redactions, issued packets, dispositions, and completed advisory outputs are database-enforced.

Unavailable sources produce partial context. Forbidden sources are counted without revealing content. Stale sources remain labeled. Failed synthesis returns the evidence manifest. Duplicate assembly and refresh operations use idempotency

keys.

Implementation-grade schema is in MASS-APP-014-V15_Migration_Reference.sql.

12. V1 Implementation

V1 supports configured source profiles, explicit assembly plans, authorization checks, reference manifests, freshness policies, contradictions, redaction, completeness and confidence indicators, immutable snapshots, executive/operational packets, and governed NOVA/POPS synthesis.

13. Future Evolution

Additional application gateways, richer temporal synthesis, privacy-preserving external federation, and expanded contradiction analysis attach through existing gateway and snapshot contracts. No future capability may create a universal shadow database.

14. Constitutional Boundary Statement

V15 assembles context; it does not own, alter, approve, or execute source truth. It cannot override permissions, resolve factual or policy conflict without human authority, or represent incomplete information as certainty.